Robert M. Gower

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modern optimization **adposure** **stochastic optimization** **machine learning** **randomized numerical linear algebra**

Profile

I work on the mathematics of modern optimization: why methods such as Adam, Muon, Polyak step-sizes, and non-Euclidean gradient descent work, fail, and sometimes come up with new methods. My current research sits between optimization theory and machine-learning practice, with a focus on simple algorithms that exploit structure. Previously, I worked on randomized numerical linear algebra, automatic differentiation, and quasi-Newton methods.

Appointments and Education

11/2021–present	**Research Scientist** , Center for Computational Mathematics, Flatiron Institute New York, NY
06/2021–11/2021	**Visiting Scientist** , Google Research New York, NY
09/2017–06/2021	**Assistant Professor in Machine Learning** , Telecom Paris, Institut Polytechnique de Paris Paris, France
01/2020–08/2020	**Visiting Scientist** , Facebook Research New York, NY
08/2016–09/2017	**Laureate Fellow** , Fondation Sciences Mathematiques de Paris, ENS and Inria Paris, France
09/2012–06/2016	**Ph.D.** , Optimization and Operations Research, University of Edinburgh Thesis: *Sketch and Project: Randomized Iterative Methods for Linear Systems and Inverting Matrices*
03/2009–04/2011	**M.Sc.** , Applied Mathematics, University of Campinas Thesis: *Hessian Matrices via Automatic Differentiation*
03/2005–03/2009	**B.Sc.** , Applied and Computational Mathematics, University of Campinas Brazil

Selected Recent News

2026	**The Polar Express** received the ICLR 2026 Outstanding Paper Honorable Mention and was presented as an oral.
2026	Upcoming talk at the SIAM Conference on Optimization (OP26), June 2–5, 2026.
2025	**In Search of Adam's Secret Sauce** was an oral presentation at NeurIPS 2025.

Prizes, Honors, and Funding

- ICLR 2026 Outstanding Paper Honorable Mention for *The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm.*
- NeurIPS 2025 Top Reviewer.
- ICML 2021 Best Reviewer, Top 10%.

- NeurIPS 2020 Top 10% reviewer.
- Thomas Jefferson Fund grant, “Matrix equations for big data compression”, $20k, 2019.
- CIFRE Facebook Research funding, $72k, 2019.
- ICML 2019 Top 5% reviewer.
- LabEx Mathematique Hadamard funding, “Scalable stochastic variance reduced gradient methods”, 8k EUR, 2018.
- Leslie Fox Prize for Numerical Analysis, 2nd place, 2017.
- Fondation Sciences Mathematiques de Paris Fellowship at Inria/ENS, 122k EUR, 2016.
- Best Talk Prize, Irish SIAM student meeting, 2014.
- University of Edinburgh Ph.D. scholarship, 85.1k GBP, 2012.

People and Mentoring

- Nidham Gazagnadou, Ph.D. student, 2018–2021; now Research Scientist at Sony AI.
- Rui Yuan, Ph.D. student, 2019–2023; now AI Research Scientist at Stellantis.
- Samy Jelassi, intern, 2017; Ph.D. at Princeton.
- Si Yi Meng (Cathy), intern, 2022 and 2023.
- Fabian Schaipp, guest researcher, 2022 and 2023.
- Slavomir Hanzely, intern, 2022; now Researcher at CISPA.
- Aaron Mishkin, intern, 2023; now Postdoc at EPFL.
- David Persson, Flatiron Fellow, 2024–present.
- Michael Crawshaw, intern, 2025; joining Flatiron as a Fellow in 2026.
- Tetiana Parshakova, Flatiron Fellow, 2025–present.

Recorded Talks

2026	**The Polar Express**. Instituto de Matematica Pura e Aplicada	recording/link
2024	**New viewpoints, variants and convergence theory for stochastic Polyak step-sizes**. MTL MLOpt	recording/link
2019	**SGD: General Analysis and Improved Rates**. ICML oral presentation	recording/link

Selected Invited Talks

Over 30 talks	in international conferences, seminars and workshops including.
2019	Parietal team seminar, Inria, Saclay
	International Conference on Continuous Optimization (ICCOPT), invited session, Berlin.
	Journée Statistique et Apprentissage, L’Institut des Hautes Études Scientifiques.
2018	The 21st International Conference on Artificial Intelligence and Statistics (AISTATS), **semi plenary presentation** and two posters, Lanzarote, Canary Islands
	Séminaire Parisien d’Optimisation, Institut Henri Poincaré
	International Symposium of Mathematical Optimization, invited session on Fast Converging Stochastic Optimization Algorithms, Bordeaux
	Seminar for the Machine Learning and Optimization group, EPFL

2017 PGMO days, EDF'Lab Paris-Saclay

Thirty-first Conference on Neural Information Processing Systems (Neurips), Poster, Vancouver.

Optimization 2017, Faculdade de Ciências of the Universidade de Lisboa, invited talk, Lisboa

The 27th Biennial Numerical Analysis Conference, Randomized Iterative Methods for Computing the Pseudoinverse, University of Strathclyde, Glasgow

18th IMA Leslie Fox Prize talk in Numerical Analysis, awarded 2nd place, University of Strathclyde, Glasgow

France/Japan Machine Learning Workshop, Ecóle Normale Supérieure, Paris

Optimization, machine learning, and pluri-disciplinarity workshop, Paris, Inria Grenoble Rhone-Alpes, September 21-22.

Probabilistic Numerics seminar, Max-Planck-Institute Tubingen

Operations Research seminar, Center for Operations Research and Econometrics, Université catholique de Louvain

2016 Computational Mathematics and Applications seminar, Rutherford Appleton Laboratory

Télécom ParisTech, Machine Learning and Statistics seminar, Paris

International Conference on Continuous Optimization, Tokyo

5th IMA Conference on Numerical Linear Algebra and Optimization, University of Birmingham
International Conference on Machine Learning (**ICML**), New York

Cambridge Image Analysis group seminar, University of Cambridge
Randomized iterative methods for linear systems

Inria, SIERRA team seminar, Paris, *Randomized iterative methods for linear systems and inverting matrices*

2015 Distributed machine learning and optimization, Alan Turing Scoping Workshop, Edinburgh, UK
Distributed Randomized Iterative Methods for Linear Systems

The International Symposium on Mathematical Programming, Pittsburgh, USA.
Action constrained quasi-Newton methods

Optimization and Big Data 2015, Edinburgh, UK
Randomized iterative methods for linear systems and inverting matrices

2014 Irish Applied Mathematics Research Students' Meeting, Galway, Ireland.
Hunting inverses of matrix fields
[**Best Talk Prize**]

Postgraduate Mathematics Colloquium, Edinburgh, UK
The history of optimization in blood and booze

Cambridge Mathematics Society Research in the UK Afternoon, Cambridge, UK
Unconstrained optimization methods

European Workshop on Advances in Continuous Optimization, Perpignan.
Generalizations of the quasi-Newton methods using an image constraint

2013 International Conference on Continuous Optimization, Lisbon, Portugal
Third order methods and third order derivatives

EURO-INFORMS Joint International Meeting, Rome, Italy
Third order methods using slices of the tensor and AD developments

2011 28th Brazilian Colloquium on Mathematics, IMPA, Rio de Janeiro, Brazil
Automatic differentiation of Hessian matrices

Teaching

I have given over 360 hours of lectures on optimization, numerical analysis, and machine learning. Prior to that, as a graduate student at the University of Edinburgh, I gave over 350 hours of tutorials and lectures.

- 2019–2020: *Machine Learning in High Dimensions*, graduate course, Telecom Paris.
- 2017–2020: *Optimization and Numerical Analysis*, undergraduate course, Telecom Paris.
- 2017–2020: *Optimization for Data Science*, Data Science Master's, Institut Polytechnique de Paris.
- 2019: *Stochastic Optimization for Machine Learning*, African Master's of Machine Intelligence at AIMS, Rwanda.
- 2017–2019: Summer school lectures on stochastic optimization for machine learning at Yerevan State University and the University of Novi Sad.
- 2010–2015: Over 350 hours of tutorials and lectures at the University of Edinburgh, including linear algebra, calculus, financial mathematics, operations research, optimization theory, proofs, and problem solving.

Service

Editorial work and reviewing for:

Action Editor, *Transactions of Machine Learning Research* (TMLR)

Editorial Board Reviewers, *Journal of Machine Learning Research* (JMLR)

Area Chair / Meta-Reviewer, *International Conference on Machine Learning* (ICML), 2025

Area Chair, *Conference on Neural Information Processing Systems* (NeurIPS), 2021 and 2022

Senior Program Committee, *International Joint Conference on Artificial Intelligence* (IJCAI), 2020

Reviewer, *Conference on Neural Information Processing Systems* (NeurIPS)

International Conference on Machine Learning (ICML)

Journal of Machine Learning Research (JMLR)

Proceedings of the IEEE

IEEE Transactions on Signal Processing

SIAM Journal on Scientific Computing (SISC)

SIAM Journal on Optimization (SIOPT)

Optimization Methods and Software, Taylor & Francis

Computational and Applied Mathematics, Springer

Mathematical Programming, Springer Computation

BIT Numerical Mathematics, Springer

Numerical Algorithms, Springer

Publications

All entries below are generated directly from `../rob_papers.bib`.

[1] N. Amsel, D. Persson, C. Musco, and R. M. Gower. "The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm". In: *International Conference on Learning Representations*. Oral presentation. 2026. arXiv: 2505.16932 [cs.LG]. URL: https://openreview.net/forum?id=yRtgZ1K8hO.

[2] M. Crawshaw, C. Modi, M. Liu, and R. M. Gower. "An Exploration of Non-Euclidean Gradient Descent: Muon and its Many Variants". In: *International Conference on Machine Learning*. 2026. arXiv: 2510.09827 [cs.LG]. URL: https://arxiv.org/abs/2510.09827.

[3] R. Islamov, M. Crawshaw, J. Cohen, and R. M. Gower. "Non-Euclidean Gradient Descent Operates at the Edge of Stability". In: *International Conference on Machine Learning*. Spotlight presentation. 2026. arXiv: 2603.05002 [cs.LG]. URL: https://arxiv.org/abs/2603.05002.

[4] F. Schaipp, R. M. Gower, and A. B. Taylor. “Step-Size Stability in Stochastic Optimization: A Theoretical Perspective”. In: *International Conference on Machine Learning*. 2026. arXiv: 2602.09842 [math.OC]. URL: https://arxiv.org/abs/2602.09842.

[5] D. Cai, R. M. Gower, D. M. Blei, and L. K. Saul. “Fisher Meets Feynman: Score-Based Variational Inference with a Product of Experts”. In: *Advances in Neural Information Processing Systems*. Spotlight presentation. 2025. arXiv: 2510.21598 [stat.ML]. URL: https://arxiv.org/abs/2510.21598.

[6] R. M. Gower, G. Garrigos, N. Loizou, D. Oikonomou, K. Mishchenko, and F. Schaipp. *Analysis of an Idealized Stochastic Polyak Method and its Application to Black-Box Model Distillation*. 2025. arXiv: 2504.01898 [cs.LG]. URL: https://arxiv.org/abs/2504.01898.

[7] A. Mishkin, A. Bietti, and R. M. Gower. “Level Set Teleportation: An Optimization Perspective”. In: *Proceedings of The 28th International Conference on Artificial Intelligence and Statistics*. Vol. 258. Proceedings of Machine Learning Research. PMLR, 2025, pp. 5059–5067. arXiv: 2403.03362. URL: https://proceedings.mlr.press/v258/mishkin25a.html.

[8] A. Orvieto and R. M. Gower. “In Search of Adam’s Secret Sauce”. In: *Advances in Neural Information Processing Systems*. Oral presentation. 2025. arXiv: 2505.21829 [cs.LG]. URL: https://openreview.net/forum?id=CH72XyZs4y.

[9] Y.-M. Pun, M. Buchholz, and R. M. Gower. *Schedulers for Schedule-free: Theoretically Inspired Hyperparameters*. 2025. arXiv: 2511.07767 [cs.LG]. URL: https://arxiv.org/abs/2511.07767.

[10] F. Schaipp, G. Garrigos, U. Şimşekli, and R. M. Gower. “Tracking the Median of Gradients with a Stochastic Proximal Point Method”. In: *Transactions on Machine Learning Research* (2025). arXiv: 2402.12828 [cs.LG]. URL: https://openreview.net/forum?id=WMxLEgYGxu.

[11] D. Cai, C. Modi, C. C. Margossian, R. M. Gower, D. M. Blei, and L. K. Saul. “EigenVI: Score-Based Variational Inference with Orthogonal Function Expansions”. In: *Advances in Neural Information Processing Systems*. Vol. 37. Spotlight presentation. 2024. arXiv: 2410.24054. URL: https://openreview.net/forum?id=thUf6ZBlPp.

[12] D. Cai, C. Modi, L. Pillaud-Vivien, C. Margossian, R. M. Gower, D. M. Blei, and L. K. Saul. “Batch and Match: Black-Box Variational Inference with a Score-Based Divergence”. In: *Proceedings of the 41st International Conference on Machine Learning*. Vol. 235. Proceedings of Machine Learning Research. Spotlight presentation. PMLR, 2024, pp. 5258–5297. arXiv: 2402.14758. URL: https://proceedings.mlr.press/v235/cai24b.html.

[13] R. M. Gower, D. A. Lorenz, and M. Winkler. “A Bregman-Kaczmarz Method for Nonlinear Systems of Equations”. In: *Computational Optimization and Applications* 87.3 (2024), pp. 1059–1098. DOI: 10.1007/s10589-023-00541-9. arXiv: 2303.08549. URL: https://doi.org/10.1007/s10589-023-00541-9.

[14] Y. Li, R. Yuan, C. Fan, M. Schmidt, S. Horváth, R. M. Gower, and M. Takáč. *Enhancing Policy Gradient with the Polyak Step-Size Adaptation*. 2024. arXiv: 2404.07525 [cs.LG]. URL: https://arxiv.org/abs/2404.07525.

[15] A. Mishkin, A. Khaled, Y. Wang, A. Defazio, and R. M. Gower. “Directional Smoothness and Gradient Methods: Convergence and Adaptivity”. In: *Advances in Neural Information Processing Systems*. Vol. 37. 2024. arXiv: 2403.04081. URL: https://proceedings.neurips.cc/paper_files/paper/2024/hash/1ac83203e88eb6cf6b30642f0239b932-Abstract-Conference.html.

[16] F. Schaipp, R. Ohana, M. Eickenberg, A. Defazio, and R. M. Gower. “MoMo: Momentum Models for Adaptive Learning Rates”. In: *Proceedings of the 41st International Conference on Machine Learning*. Vol. 235. Proceedings of Machine Learning Research. PMLR, 2024, pp. 43542–43570. arXiv: 2305.07583. URL: https://proceedings.mlr.press/v235/schaipp24a.html.

[17] B. Zhao, R. M. Gower, R. Walters, and R. Yu. “Improving Convergence and Generalization Using Parameter Symmetries”. In: *International Conference on Learning Representations*. 2024. arXiv: 2305.13404 [cs.LG]. URL: https://openreview.net/forum?id=qXe7dE7Fq8M.

[18] J. Domke, R. M. Gower, and G. Garrigos. “Provable Convergence Guarantees for Black-Box Variational Inference”. In: *Advances in Neural Information Processing Systems*. Vol. 36. 2023. URL: https://proceedings.neurips.cc/paper_files/paper/2023/hash/d0bcff6425bbf850ec87d5327a965db9-Abstract-Conference.html.

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[20] A. Khaled, O. Sebbouh, N. Loizou, R. M. Gower, and P. Richtárik. “Unified Analysis of Stochastic Gradient Methods for Composite Convex and Smooth Optimization”. In: *Journal of Optimization Theory and Applications* 199.2 (2023), pp. 499–540. DOI: 10.1007/s10957-023-02297-y. arXiv: 2006.11573. URL: https://doi.org/10.1007/s10957-023-02297-y.

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[22] S. Y. Meng and R. M. Gower. "A Model-Based Method for Minimizing CVaR and Beyond". In: *Proceedings of the 40th International Conference on Machine Learning*. Vol. 202. Proceedings of Machine Learning Research. PMLR, 2023, pp. 24436–24456. URL: https://proceedings.mlr.press/v202/meng23a.html.

[23] C. Modi, C. Margossian, Y. Yao, R. M. Gower, D. M. Blei, and L. K. Saul. *Variational Inference with Gaussian Score Matching*. 2023. arXiv: 2307.07849 [stat.ML]. URL: https://arxiv.org/abs/2307.07849.

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[26] J. Chen, R. Yuan, G. Garrigos, and R. M. Gower. "SAN: Stochastic Average Newton Algorithm for Minimizing Finite Sums". In: *Proceedings of The 25th International Conference on Artificial Intelligence and Statistics*. Vol. 151. Proceedings of Machine Learning Research. PMLR, 2022, pp. 279–318. arXiv: 2106.10520. URL: https://proceedings.mlr.press/v151/chen22b.html.

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